UNIT - IV

POLYMERS

Polymer (Greek poly = many; mers = units or parts)**:** A polymer is a macromolecule (giant molecule of high molecular mass) built-up by the linking together of a large number of simple molecules (monomers).
Example: Polythene is formed by linking a large number of ethene molecules together.

$$\underset{\text{ethene (monomer)}}{n\,CH_2{=}CH_2} \longrightarrow \underset{\text{polythene (polymer)}}{\left(CH_2\text{-}CH_2\right)_n}$$

Monomer: A monomer is a simple molecule having two or more bonding sites through which each can link to other monomers to form a polymer chain. Monomers are often called building blocks of a polymer chain.

Functionality: The total number of functional groups or bonding sites present in a monomer molecule is called the functionality of the monomer. For a substance to act as a monomer, it must have at least two reactive sites or bonding sites (*bifunctional*). Functionality of some commercial monomers:

Sl. No.	Monomers	Functional group/Reactive centres	Functionality
1.	Vinyl compounds	Two reactive centres at the double bond	Two
2.	Ethylene glycol	Two alcoholic -OH groups	Two
3.	Adipic acid	Two –COOH groups	Two
4.	Hexamethylene diamine	Two $-NH_2$ groups	Two
5.	Phenol	Three active centres at ortho and para positions	Three
6.	Formaldehyde	Two active centres	Two
7.	Bisphenol-A	Two phenolic –OH groups	Two

Polymerization: The chemical process by which the monomers (low molecular weight) are converted into polymers (high molecular weight), at a particular temperature and pressure, is called polymerization. All polymerization reactions need suitable initiators.

$$\text{Monomers + Initiator} \xrightarrow[\text{Pressure}]{\text{Temperature}} \text{Polymer}$$

Degree of polymerization (DP): The number of repeating units (*n*) in a polymer chain is called degree of polymerization. There may be hundreds or thousands or tens of thousands or more monomer molecules linked together in a polymer molecule.

$$\underset{\text{ethene}}{n\,CH_2{=}CH_2} \longrightarrow \underset{\text{polythene}}{\left(CH_2\text{-}CH_2\right)_n}$$

where '*n*' is the degree of polymerization.

High polymers: The polymers with high degree of polymerization (10,000 to 1,00,000) and high molecular masses (10,000 to 10,00,000) are called high polymers.

Classification of Polymers

(a) Natural and Synthetic polymers (*based on origin*): The polymers which are obtained from natural sources such as plants and animals are called natural polymers.
E.g. Cotton, Wool, Silk, Starch, Cellulose, Proteins, Natural rubber, Leather etc.
The polymers which are synthesized from simple molecules are called synthetic polymers. E.g. Polyethylene, Polystyrene, Teflon, PVC, Plexiglass etc.

(b) Homopolymers, Copolymers and Linear, Branched and Crosslinked polymers (*based on structure*): A polymer may consist of identical monomers or monomers of different chemical structures and accordingly they are, called **homopolymers** and **copolymers** respectively.

-M-M-M-M-M-M-M-M- Homopolymer (Eg. Polyethylene, PVC etc.)
$-M_1-M_2-M_1-M_2-M_1-M_2-$ Copolymer (eg. Nylon 66, Polycarbonates etc.)

Based on the way in which repeat units are linked, they are named as Linear, Branched and Cross-linked polymers.
A **Linear polymer** is one in which each monomer unit is linked only to two neighbouring units.

-M-M-M-M-M-M-M-M- Linear homopolymer (eg. Polyethylene)
$-M_1-M_2-M_1-M_2-M_1-M_2-M_1-$ Linear copolymer (eg. Nylon 66)

Linear copolymers in which a block of one repeat unit is followed by a block of another repeat unit consecutively is called a **block copolymer**.

$-M_1-M_1-M_1-M_1-M_2-M_2-M_2-M_2-$

A **branched chain polymer** is one in which long linear chain may branched out forming a branched structure.

```
-M-M-M-M-M-M-M-M-
      |
      M
      |
      M
      |
      M
      |
```

Branched chain homopolymer

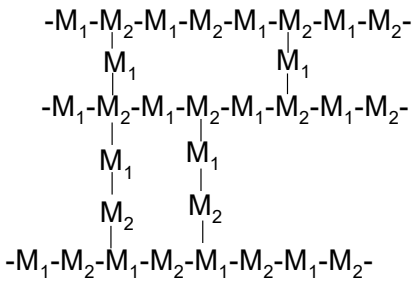
```
-M1-M2-M1-M2-M1-M2-M1-M2-
      |
      M2
      |
      M1
      |
      M2
      |
```

Branched chain copolymer

Branched copolymers with one kind of monomers in their main chain and another kind of monomers in their side chain are called **graft copolymers.**

```
-M1-M1-M1-M1-M1-M1-M1-M1-
      |
      M2
      |
      M2
      |
      M2
      |
```

Cross linked polymer is formed when linear molecules under certain conditions are linked to neighbouring ones in such a way that it results in the formation of a large three-dimensional structure.



(eg. phenol-formaldehyde and urea-formaldehyde)

(c) Addition and condensation polymers (*based on method of formation*): Addition polymers are those formed by a process of self-addition without elimination of by-products. Example – polyethylene, polypropylene etc.
Condensation polymers are formed by condensation reaction i.e., reaction between two or more monomer molecules with elimination of simple molecules like water, ammonia and hydrogen chloride etc. Example- phenol-formaldehyde resins and synthetic fibres such as polyesters and polyamides.

(d) Thermoplastic and Thermosetting Polymers (*based on response to heat*): Some polymers, on the application of heat and pressure, initially become soft, flexible rubbery matter and undergo deformation. On further heating beyond melting point, they melt and flow. Such property of polymer is called plasticity. On cooling they return to their original state. Thermoplastics exhibit this property. Thermoplastics are linear, stereoregular polymers. The polymer chains are closely packed and held by secondary forces such as vander waal, hydrogen bonding and dipolar interaction. Such polymers when heated, the chains acquire sufficient energy and overcome these inter chain attractive forces. They attain molecular mobility and flow like viscous fluid. Eg. Polyethylene, polypropylene, poly vinylchloride, etc.

Thermosettings do not exhibit plasticity. Moulded thermosettings have three-dimensional structure. All monomer units are held together through strong primary covalent bonds throughout the structure. On strong heating, charring occurs instead of deformation. Therefore, thermosettings do not undergo reversible plastic deformation. Eg. Bakelite, cross-linked polyurethane, silicone resins, etc.

Molecular weight of polymers

A simple molecule has a fixed molecular weight. In contrast, a polymer comprises molecules of different molecular weights and hence, its molecular weight is expressed in terms of an 'average' value. E.g., ethylene gas, which is a low molecular weight compound, each of its molecules has the same chemical structure and hence, a fixed molecular weight of 28. But upon polymerization, it forms polyethylene and we encounter an indefinite chemical structure

of --($CH_2 – CH_2$ -)$_n$- where *n* can change its value from one polyethylene molecule to another present in the same polymer sample.

When ethylene is polymerized to form polyethylene, a number of polymer chains start growing at any instant, but all of them do not get terminated after growing to the same size. The chain termination is a random process and hence, each polymer molecule formed can have a different number of monomer units and thus different molecular weights. In other words, molecular weight of the given polymer sample depends upon the degree of polymerisation of various polymeric chains present.

The molecular weight of the synthetic polymer can be taken statistically and expressed as average of the molecular weights contributed by the individual molecules. So the molecular weight of a polymer can be expressed by two most and experimentally verifiable methods of averaging viz., (i) Number-average and (ii) Weight-average.

Number-average molecular weight: Number-average molecular mass of a polymer can be defined as the total mass of all the molecules in a polymer sample divided by the total number of molecules present.

$$\overline{M_n} = \frac{n_1M_1}{\sum n_i} + \frac{n_2M_2}{\sum n_i} + \frac{n_3M_3}{\sum n_i} + \frac{n_4M_4}{\sum n_i} + \ldots\ldots\ldots\ldots\ldots\ldots + \frac{n_iM_i}{\sum n_i}$$

$$\overline{M_n} = \frac{\sum n_i M_i}{\sum n_i}$$

Weight-average molecular weight: The sum of the fractional masses that each molecule contributes to the average according to the ratio of its mass to that of the whole sample.

$$\overline{M_w} = \frac{n_1M_1^2}{\sum n_iM_i} + \frac{n_2M_2^2}{\sum n_iM_i} + \frac{n_3M_3^2}{\sum n_iM_i} + \frac{n_4M_4^2}{\sum n_iM_i} + \ldots\ldots\ldots\ldots + \frac{n_iM_i^2}{\sum n_iM_i}$$

$$\overline{M_w} = \frac{\sum n_i M_i^2}{\sum n_iM_i}$$

Polydispersity Index (PDI): PDI is the measure of the distribution of molecular weights in a polymer sample. It is expressed as the ratio of weight average molecular weight and number average molecular weight.

$$\boldsymbol{PDI = \frac{\overline{M_w}}{\overline{M_n}}}$$

A PDI equal to one indicates the polymer to be monodisperse with all the polymer chains of the same length. Most of the natural polymers are monodisperse. Synthetic polymers generally have PDI >1 and they are less homogeneous and polydisperse.

Example:

A polymer sample contain 50% molecules having a molecular weight of 2000, 40% molecules having a molecular weight of 3000 and 10% molecules having a molecular weight of 4000. Calculate its number average and weight average molecular weight and polydispersity index.

Solution:

The number average molecular weight in given by $\overline{M}_n = \frac{\Sigma\, n_i Mi}{\Sigma n_i}$

$$\overline{M}_n = \frac{0.5 \times 2000 + 0.4 \times 3000 + 0.1 \times 4000}{0.5 + 0.4 + 0.1}$$

$$\overline{M}_n = 2600$$

The weight average molecular weight is $\overline{M}_w = \frac{\Sigma n_i\, M_i^2}{\Sigma n_i M_i}$

$$\overline{M}_w = \frac{0.5 \times (2000)^2 + 0.4 \times (3000)^2 + 0.1 \times (4000)^2}{0.5 \times 2000 + 0.4 \times 3000 + 0.1 \times 4000}$$

$$\overline{M}_w = 2769.23$$

$$PDI = \frac{\overline{M_w}}{\overline{M_n}} = \frac{2769.23}{2600} = 1.06$$

Polymer Composites

Structural materials are required to possess properties such as, (i) low density (ii) high strength and stiffness (iii) abrasion and impact resistance and (iv) corrosion resistance. No single metal, alloy, ceramic or polymeric material is known which can offer the combination of aforesaid properties. This is because a strong material is relatively dense and an increase in stiffness generally results in a decrease in impact strength. The search for materials possessing the above properties has led to the development of composite materials.

Definition: A combination of two or more distinct components to form a new class of material suitable for structural applications is referred to as composite materials. When one of the components is a polymer, the resulting composite is called a polymer composite.

While each component retains its identity, the composite material displays macroscopic properties superior to its parent constituents, particularly in terms of mechanical properties.

Polymer composites are generally made of two components, namely (i) matrix and (ii) fibre (Fig. 1). The fibre is embeded in the matrix in order to make the matrix stronger. The matrix is usually a thermoset material such as epoxy resin or a polyamide and it holds the fibres together. Fibre is most often glass but sometimes may be a carbon fibre, polyethylene or Kevlar**.**

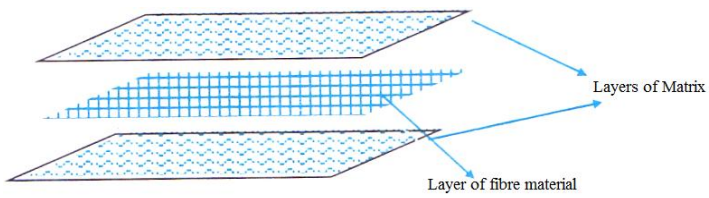
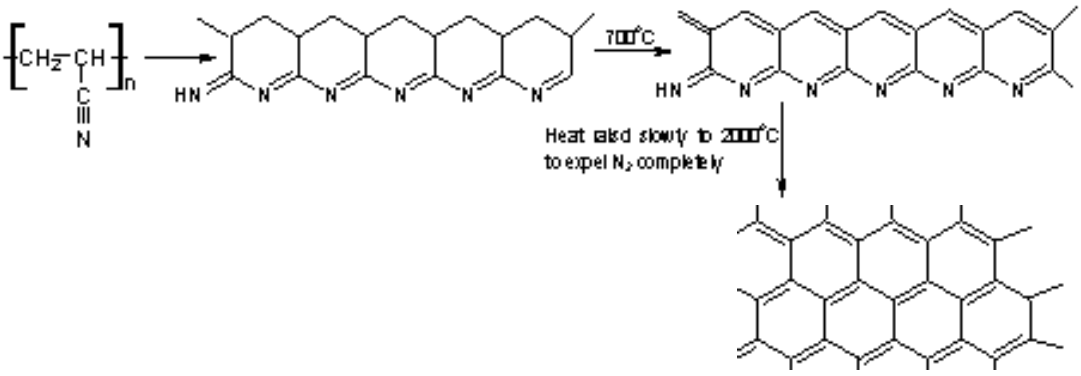


Fig. 1 Polymer composite with the matrix and fibre components

Fibre-reinforced composites are strong and light. They are often stronger than steel, but weigh much less and hence these composites can be used to make automobiles lighter, achieving more fuel efficiency and less pollution.

Carbon fibre: Carbon fibre is a synthetic fibre consisting of very thin strands of carbon, even thinner than human hair. When bound together with plastic polymer resin with heat, pressure or vacuum, a composite material is formed that is both strong and light weight. The strength of carbon fibre depends on the weave. The more complex the weave the more durable the composite will be.

Synthesis: Carbon fibre is prepared by heating polyacrylonitrile (PAN) carefully and slowly. During this treatment, the polymer cyclizes through the cyano groups to form a polycylic chain. The resulting solid is heated gradually so that hydrogen is expelled and rings become aromatic. It is then roasted at 600-700^{0}C when the adjacent chains join together losing more hydrogen gas. Then the temperature is gradually raised to 2000^{0}C to get wider ribbon-like mass. The temperature is maintained around 2000^{0}C till all the nitrogen is expelled leaving behind wider ribbon-like pure carbon fibre in the graphite form.



Properties: (i) Carbon fibre reinforced composites are very strong and are often stronger than steel but lighter (ii) Good corrosion resistance (iii) Very low coefficient of thermal expansion (iv) Low impact resistance.

Applications: (i) In aerospace and automotive fields, as well as in sailboats (ii) In modern bicycles and motorcycles (iii) In consumer goods such as laptops, tripods, finishing rods, golf clubs, tennis rackets etc.

Conducting Polymers

Conducting polymers are polymers that can conduct electricity, unlike typical polymers which are insulators. An organic polymer, with linear structure and extensive conjugation (alternate single and double bonds) in its polymeric back bone, can be converted into a conducting polymer by doping it with an oxidizing or a reducing agent.

E.g., Polyacetylene, Polythiophene, Polyaniline, etc.

Conducting mechanism of Polyacetylene

Polyacetylene is the simplest conjugated polymer. The neutral polymer would have band model identical to that of an insulator. The polymer may be transformed into a conductor by doping it with an electron donor (reductive dopant such a sodium naphthalide) or an electron acceptor (oxidizing dopant such as iodine, $FeCl_3$). Doping introduces delocalized positive or

negative charges on the polymer backbone, thereby increasing its electrical conductivity to levels comparable with metals.

(i) Oxidative doping (p-doping): In this process, polyacetylene is partially oxidized using suitable oxidizing agent such as iodine in carbon tetrachloride (Fig. 2). This leads to the removal of an electron from π-system of the backbone producing free radical and a positive charge. The positive charges are compensated by I^-_3 anions. The resulting combination of radical cation is referred to as polaron. This would create new localized electronic states in the forbidden gap being occupied by a single unpaired electron. Removal of second electron creates bipolaron state which undergo radical recombination. Upon further oxidation, the free radical of the polaron is removed, creating charge carriers (positive charges) along the polymer chain called as soliton. The delocalized positive charges are the current carriers for conduction. When doping is high, several charged solitons form soliton band. This can later merge with edges of valence and conduction bands exhibiting conductivity.

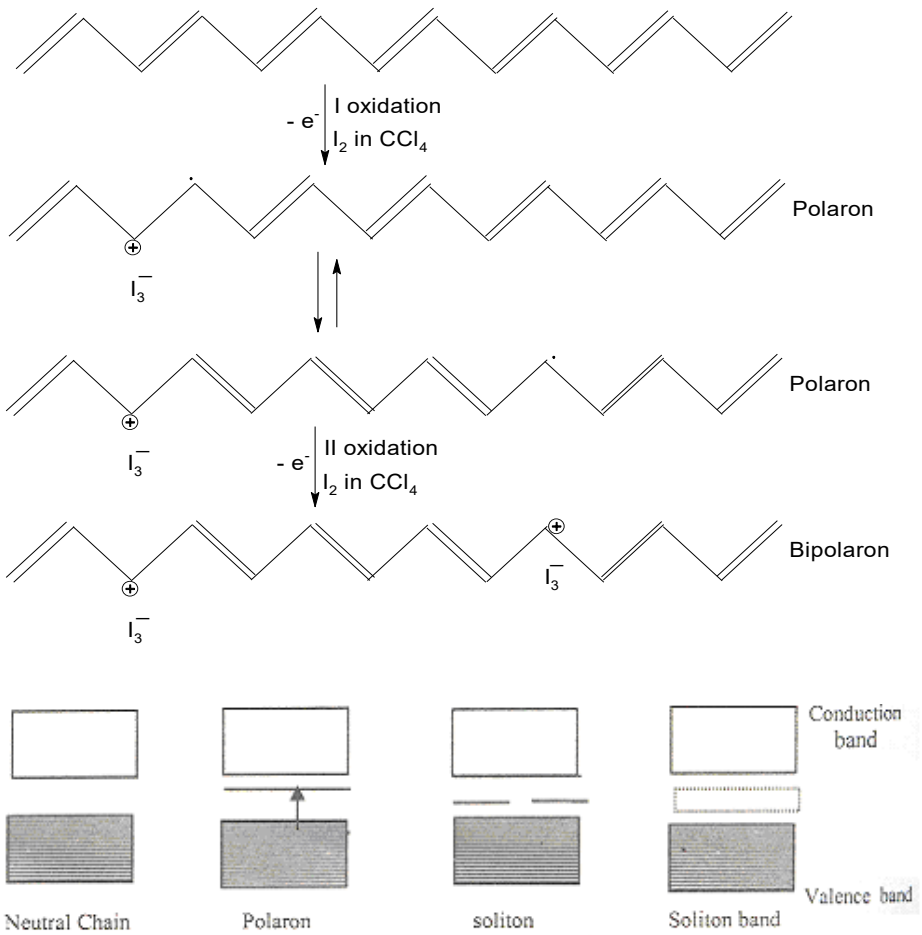


Fig. 2 Mode of conduction in oxidative doping of polyacetylene and the related band diagram

(ii) Reductive doping (n-doping): In this process, polyacetylene is partially reduced using reducing agent such as sodium naphthalide in tetrahydrofuran (Fig. 3). Here, the reducing agent donates an electron leading to the formation of a polaron (radical anion). Addition of a second electron, results in a bipolaron which on further reduction, undergo radical recombination and segregate to form solitons. Here, the negative charges on the polymer back bone are the current

carriers for conduction. As observed in oxidative doping, the electrons added by the reducing agent occupy the intermediate state within the band gap to eventually form a soliton band between the valence band and the conduction band. The soliton band allows facile jump of electrons from the valence band in to the conduction band, and hence the doped polyacetylene has higher conductivity than the undoped polymer.

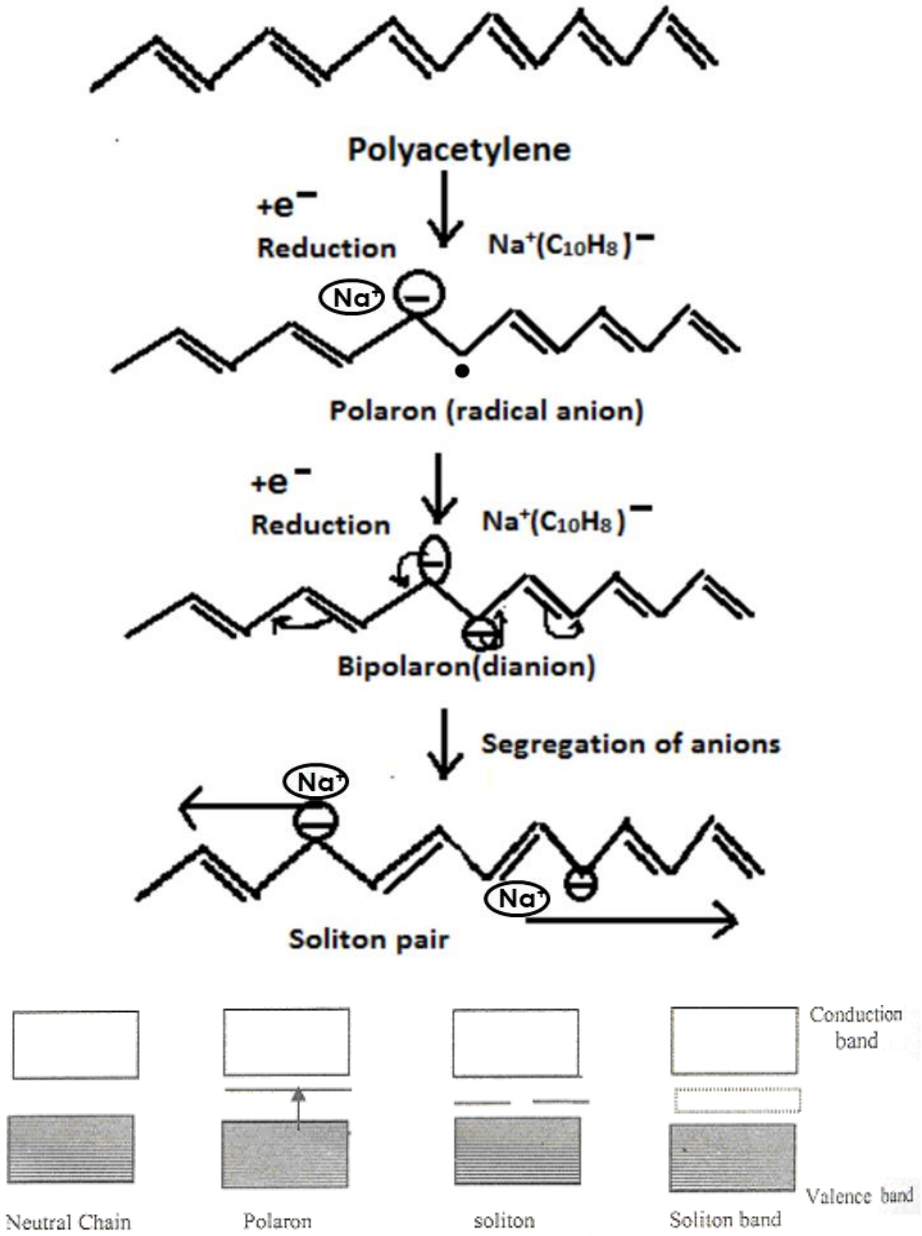


Fig. 3 Mode of conduction in reductive doping of polyacetylene and the related band diagram

Applications of conducting polymers

1. Used as electrode material for commercial re-chargeable batteries
2. As a conductive track on printed circuit boards
3. As a sensor material for humidity, gas and biomolecules
4. In light emitting diodes
5. In electrochromic display windows
6. In information storage devices

NANOMATERIALS

Nanomaterials are materials which have at least one of their dimensions in the nanometer range (1 nm = 10^{-9} m). Nanostructures include quantum dots, clusters, nanocrystals, nanowires, nanotubes and so on. The physical and chemical properties of nanomaterials can differ significantly from those of their bulk counterparts. Suitable control of the properties and response of nanostructures can lead to new devices and technologies. The nanoscience field has matured so rapidly that it is probably hard to find a segment of any technical subject where the implications of nanomaterials have not been explored at least to a preliminary extent.

Classification of nanomaterials (*based on dimension*)

The classification is based on the number of dimensions which are not confined to the nanoscale range (< 100 nm).

(i) Zero dimensional: Materials wherein all the dimensions are within the nanoscale range are called as 0-D nanomaterials. The most common representation of 0-D nanomaterials are nanoparticles.

(ii) One dimensional: Here two dimensions are at nanoscale range, another dimension is not. This leads to needle like-shaped nanomaterials. Eg. Nanotubes, nanorods, nanowires.

(iii) Two dimensional: One dimension is at nanoscale range, other two dimensions are not. They exhibit plate-like shapes. Eg. Nano coatings, nanofilms.

(iv) Three dimensional: These materials are not confined to the nanoscale in any dimension. These materials are characterized by having three arbitrarily dimensions above 100 nm. They are also called as bulk nanomaterials. Bulk nanomaterials can form a nanocrystalline structure by multiple arrangement of nano size crystals, most typically in different orientations. 3-D nanomaterials can contain dispersions of nanoparticles, bundles of nanowires/nanotubes as well as multi nanolayers.

Properties of Nanomaterials

Nanomaterials have the structural features in between of those of atoms and the bulk materials. While most microstructured materials have similar properties to the corresponding bulk materials, the properties of nanomaterials are significantly different from those of atoms and bulks materials. This is mainly due to the nanometer size of the materials which render them: (i) large fraction of surface atoms; (ii) high surface energy; (iii) spatial confinement; (iv) reduced imperfections, which do not exist in the corresponding bulk materials. The total surface area (or) the number of surface atom increases with reducing size of the particles.

(a) **Catalytic properties:** Nanomaterials have a very large surface area-to-volume ratio, exposing more active sites for reactions. Most of the times, the surface atoms are unsaturated, making them highly reactive when compared to their bulk counterparts For example, gold nanoparticles catalyze CO oxidation, while bulk gold does not. The catalytic behavior can change significantly with the size, shape, and crystal structure of the nanoparticles.

(b) **Thermal properties:** Thermal behaviour of nanomaterials are largely dependent on the particle size as well as dimension. In bulk materials, heat is transferred through the movement of phonons. In nanoscale materials, a large amount of atoms are available

on the surface with several defects, grain boundaries and interfaces. This results in phonon scattering (phonon-phonon collision) which disrupt the phonon flow and reduces the overall thermal conductivity. However, there are exceptions such as graphene monolayers (2D nanomaterial) and carbon nanotubes (1D nanomaterial). The stiff carbon-carbon bonds along the plane (axis) allows a fast directional flow of heat by minimizing phonon scattering. This results in exceptionally high thermal conductivity, exceeding that of most metals. For comparison, the thermal conductivity of graphene monolayer is 2000-5000 W/m.K and carbon nanotubes is 3000-3500 W/m.K which is much higher than that found for bulk copper (400 W/m.K).

Synthesis of Nanomaterials

There are two approaches adopted for the synthesis of nanomaterials. They are top down and bottom up (Fig. 1). The top-down approach refers to successive cutting of a bulk material to get nanoparticles. Bottom-up approach is the building up of a material from the bottom, that is, atom by atom, molecule by molecule.

Attrition or ball milling, photolithography are examples for top-down approach. Sol gel method, precipitation method, chemical vapor condensation method, hydrothermal synthesis, laser ablation are some of the examples for bottom-up approaches.

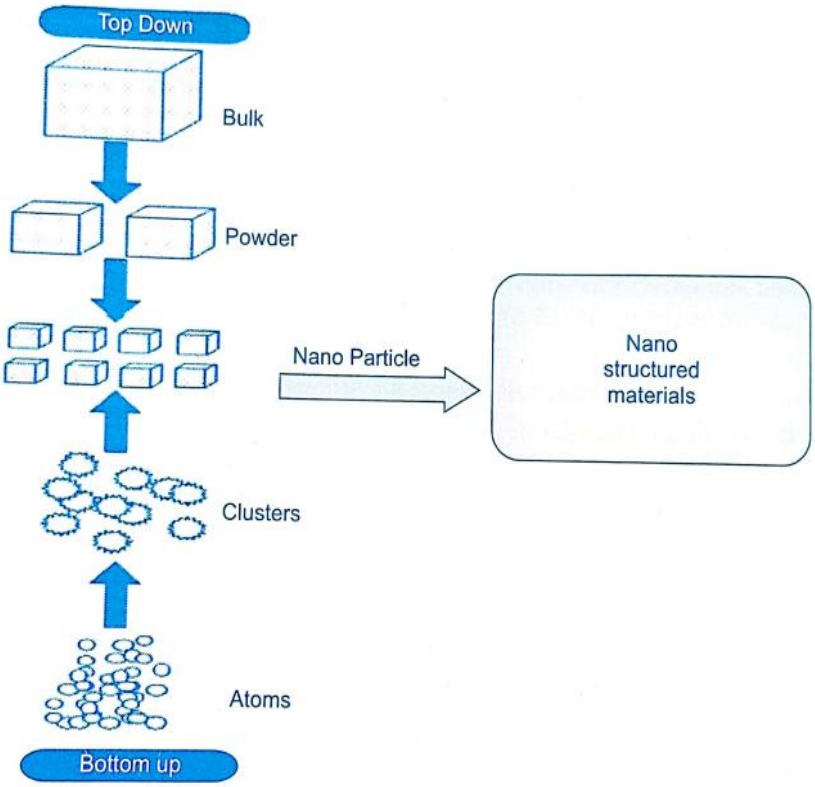


Fig. 1 Schematic representation of the top-down and bottom-up approaches for synthesis of nanomaterials

Top-down approach

Photolithography : Photolithography, also termed optical lithography or UV lithography, is a process used in fabrication of nanoscale patterns and structures on the bulk of a substrate such as silicon wafer. It uses light to transfer a geometric pattern from a photomask to a light-sensitive chemical "photoresist" applied on the substrate (Fig. 2). The key steps involved during photolithography are

(a) Cleaning

The organic or inorganic contaminations present on the substrate (silicon wafer) are removed by cleaning with hydrogen peroxide and acid formulations. The substrate is then thermally oxidized at 150°C to grow a thin layer of SiO_2 on the surface.

(b) Photoresist application and baking

The wafer is covered with photoresist by spin coating. Photoresist is a polymer solution which changes its chemical structure on exposure to UV radiation. There are two types of photo-resists: (i) positive photoresist and (ii) negative photoresist. Positive photoresists dissolve when exposed to UV light. Ex: Polymethyl methacrylate. Negative photoresists harden on exposure to UV light. Ex: hydroxyl styrene. The spin coating process results in the deposition of a uniform and thin (5-10 nm) layer of the photoresist on the substrate. The wafer is then heated slightly at 80°C ("soft bake") to remove excess solvent and improve adhesion.

(c) Mask alignment and UV exposure

A photomask (made up of transparent quartz plate) that contains the desired pattern is then placed over the photoresist. UV-light is shone through the photomask, exposing the photoresist in certain areas. The exposed areas undergo a chemical change, making them either soluble (positive photoresist) or insoluble (negative photoresist) in a developer solution. After UV exposure the photomask is removed and can be reused for other wafers.

(d) Development

The exposed or unexposed areas of the photoresist are selectively removed by immersing it in a developer solution (alkaline aqueous solutions such as tetramethyl ammonium hydroxide). The wafer is then again heated at 150°C ("hard bake") to harden the remaining photoresist and improve its resistance to etching.

(e) Etching

The exposed surface material is then etched away using chemical (strong acids) or plasma (gases) etching. The photoresist is not affected during the etching process.

(f) Photoresist removal

The remaining photoresist is stripped off chemically leaving behind the desired pattern.

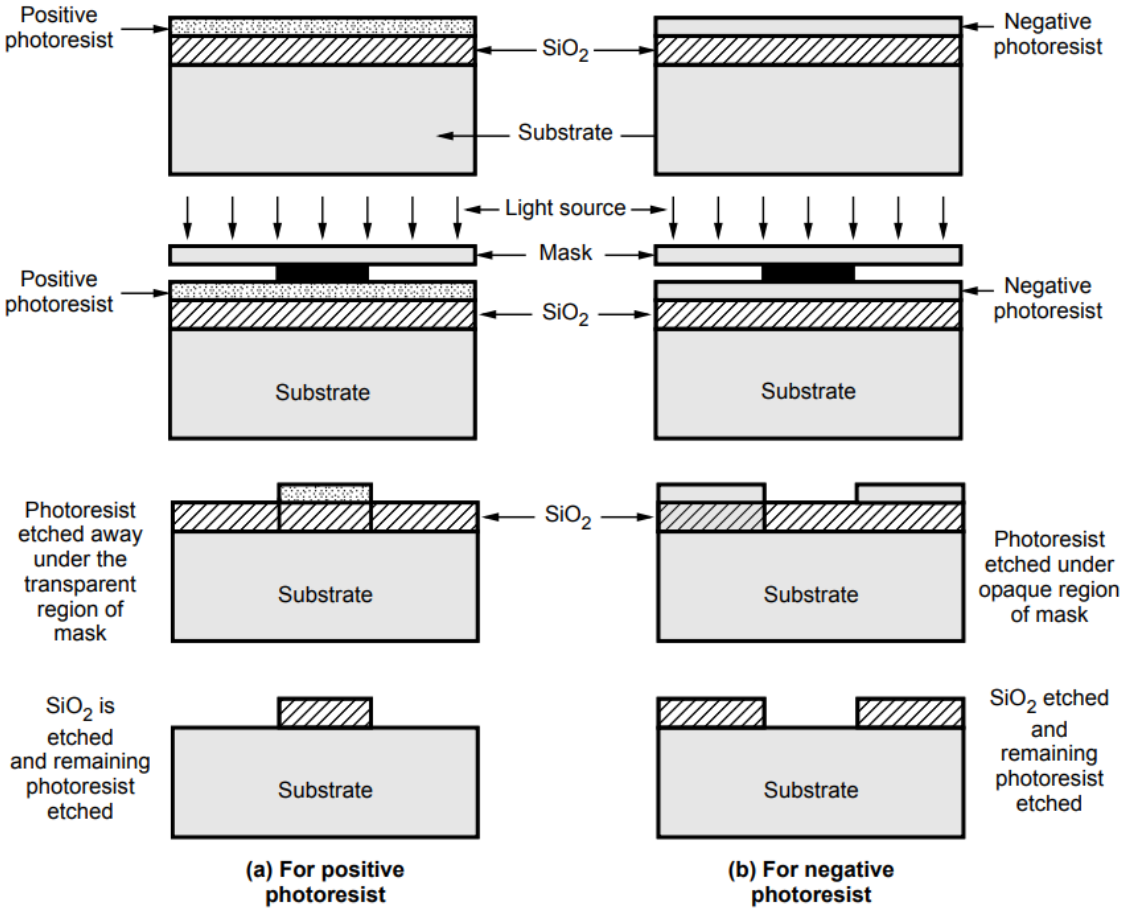
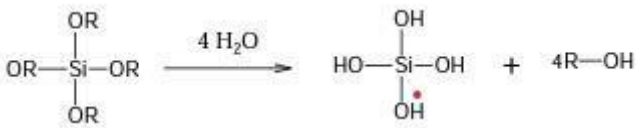


Fig. 2 Nanofabrication of silicon wafer by photolithography

Bottom-up approach

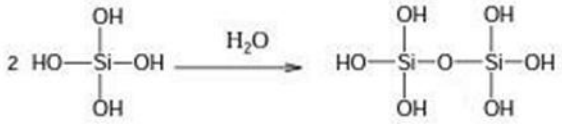
Sol-gel method: Sol-gel approach is most widely used for nanocomposite fabrication due to its simple processing steps, the flexibility of solution chemistry, the low temperature treatments and small investment in equipment. A sol is a stable suspension of colloidal solid particles in a liquid. A gel is a porous, three-dimensional, continuous solid network surrounding a liquid phase. The sol-gel process, for the development of silica aerogel, consists of following steps:

(a) Preparation of precursor solution: The desired colloidal particles are dispersed in a liquid to form a sol. The precursors generally used for the formation of sol are metal alkoxides. This is mainly because alkoxides readily react with water. A catalyst may be used to start the reaction as well as to control the pH. The first reaction to occur is the hydrolysis which converts the alkoxide (tetraethyl orthosilicate or tetramethyl orthosilicate) to alcohol (silanol).

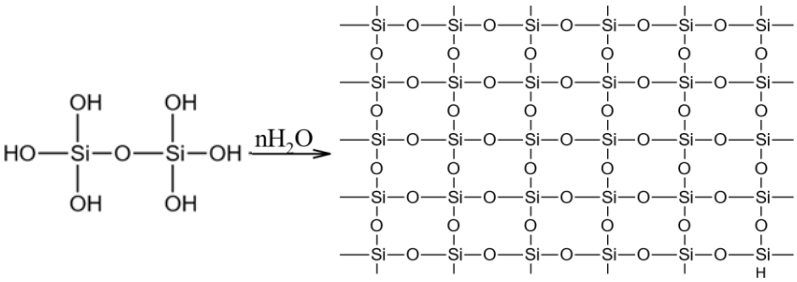


(b) Formation of gel: The particles in the sol are polymerized through the removal of the stabilizing components. This can be done by either heating the sol at a low temperature or allowing it to stand for certain duration. This results in the formation of a continuous network of [M-O-M] bonds (gel) by condensation polymerization accompanied by the liberation of

water molecules. These reactions increase the molecular weight of the oxide polymer. During the synthesis of silica aerogel, self-condensation of silanol groups take place through siloxane linkages.



(c) Aging of gel: The reaction mixture is allowed to continue polycondensation reactions until the gel transforms into a solid mass, accompanied by contraction of the gel network and expulsion of solvent from gel pores. This results in the formation a large three-dimensional structure of poly condensed silica gel with Si-O-Si linkages.



(d) Drying of the gel: Water and other volatile liquids are removed from the gel network. If isolated by thermal evaporation, the resulting nanomaterial is called xerogel. If the solvent is extracted under supercritical or near supercritical conditions, the product is an aerogel. For silica aerogel, the conditions maintained are maintained at a critical pressure of 78 bar and critical temperature of 294°C respectively. This is then calcined at temperatures up to 800°C to remove the surface bound M-OH groups followed by densification.

Synthesis of carbon nanotubes (CNT) by Chemical vapour deposition method

Chemical Vapor Deposition (CVD) technique is used to produce thin films, nanostructures, or coatings on a substrate by chemical reactions of vapor-phase precursors. Schematic diagram of the chemical vapor deposition apparatus is shown in Fig. 3.

The CVD method is commonly employed to prepare carbon nanotubes (CNT). The precursor gases are the hydrocarbons such as methane, carbon monoxide and acetylene. Initially, the quartz tube kept in the oven is heated in an inert (Ar) or reducing (H_2) atmosphere. Next, the hydrocarbon gas is introduced at high temperature (~ 720°C). At such high temperature, the hydrocarbons are broken, producing pure carbon molecules and hydrogen gas. The gaseous carbon diffuses towards the substrate, which is coated with a catalyst (usually a first-row transition metal such as Ni, Fe or Co). The metal nanoparticles act as nucleation sites where the carbon vapors bind, preferentially to form hexagonal graphene sheets. The small size of the nanoparticles directs the growth of the sheets in to cylindrical shapes rather than flat sheets. Both single-walled and multi-walled carbon nanotubes can be prepared through CVD technique. Due to the unique properties of CNTs, such as high thermal and electrical conductivity, they find applications as structural materials in thermal materials, electronics and energy devices.

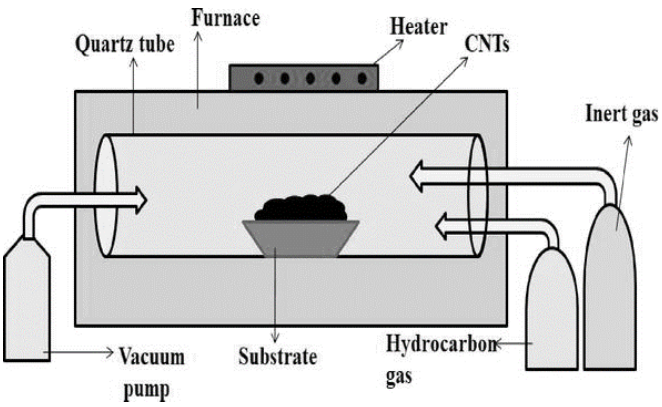


Fig. 3 Schematic diagram of the chemical vapor deposition apparatus

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